

Real-Time Upper Body Motion Tracking Using Computer Vision for Improved Human-Robot Interaction and Teleoperation

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Abstract—Upper body motion tracking mapping is crucial for robot control because it gives the machine a better understanding of how a human operator moves, allowing it to react instinctively and naturally. Most current research has focused on using wearable sensors and remote controls to enhance communication between robots and humans. However, this research aims to address the issue by embracing a non-wearable sensor-based strategy to promote more natural and spontaneous interactions between humans and robots. Moreover, A 3-DoF manipulator was also designed and developed utilizing robotics technologies. The vision system captured a human operator's upper body movements in real-time video footage. Computer vision approaches were used to extract positional and orientation information from the upper body in this setting. The system combines the MediaPipe pose model with kinematics theories to estimate the hands' position and movement in real-time. According to the experiment results, the system's overall accuracy is 94.1 (± 1.2) %, and the motion tracking system's accuracy is 96.5 (± 2.0) %.

Keywords—Assistive robotics, Control systems, Computer vision, Human-robot interaction, Upper body tracking

I. INTRODUCTION

Real-time upper body motion tracking is a fast-evolving technology that has the potential to revolutionize robotics and teleoperation. Using sophisticated, intelligent system techniques enables real-time tracking of a person's upper body motions. It is now feasible to maneuver a robot or other equipment naturally and intuitively without complicated control interfaces by precisely tracking a person's upper body movement. Real-time upper body motion tracking has enormous potential in Human-Robot Interaction (HRI) and teleoperation [1]. To create robots that can efficiently communicate with humans and help them with various activities, HRI entails the study of interactions between humans and robots [2]. On the other hand, teleoperation entails the remote control of robots or other equipment, sometimes in hazardous or inaccessible situations.

The efficacy and efficiency of the interaction or control interface can be significantly increased in both HRI and teleoperation when it is possible to follow the movement of a person's upper body precisely. For instance, real-time upper body motion tracking can allow the operator to control the robot's movements in teleoperation scenarios, such as remote

manipulation in a dangerous environment, without complicated and onerous control interfaces.

A wearable sensor-based approach to record and evaluate a person's upper body movements is essential for real-time upper body motion tracking [3]. The inertial motion trackers (IMU) are set to record various parameters of the subject's upper body. Then, the acquired data are analyzed using cutting-edge Artificial Intelligence (AI) algorithms to extract important details like joint locations and movement trajectories [4]. The real-time movement of a robot or other device is then controlled using these features.

Several applications have previously shown the promise of real-time upper body motion monitoring utilizing different techniques. For instance, real-time upper body motion tracking has been used in teleoperation to remotely control robots in hazardous settings like nuclear power plants and oil rigs [5]. Managing a robot's actions naturally and intuitively can increase safety and efficiency while lowering the possibility of human damage. It is also used in interactive gaming, allowing players to control game characters with upper body movements. This encourages fitness and physical activity while making gaming more immersive and interesting. Real-time upper body motion tracking is utilized in the healthcare industry to track patients' movements during therapeutic activities [6]. Clinicians may monitor a patient's development and offer comments on appropriate techniques while ensuring that patients carry out exercises safely and successfully by keeping track of their upper body motions.

For some sensitive circumstances like healthcare, upper body motion tracking for HRI is primarily a sensor-based technique, which is very uncomfortable [7]. All current robot control techniques rely on wearables or physical components to manage robot activities, which may not be possible or practicable depending on the surrounding environment. To address the above problem, we proposed a non-wearable sensor-based upper body motion tracking method (pose tracking algorithm using MediaPipe BlazePose model) to carry out HRI to address the issue above. We also designed and developed a 3-DoF (Degrees of Freedom) robot manipulator to test the teleoperation concept. Additionally, the system analyses the upper body's recognized positions and coordination to establish the robot arm's intended control actions.

II. LITERATURE REVIEW

A gesture-based remote-control system is an innovative approach to controlling the movement of a compliant robot manipulator. The system allows the user to control the robot manipulator through gestures made by the user's hand or body. This literature review analyzes the existing tools and methods for robot manipulators' upper body motion-tracking remote-control systems, focusing on the state-of-the-art techniques and their limitations.

A. MediaPipe Pose Landmark Model

The Google MediaPipe team developed the pose tracking model known as MediaPipe Pose. The head, torso, arms, and legs are important body joints and landmarks that the machine learning model utilizes computer vision to identify and track. To allow precise recognition and tracking of body landmarks, the model is built on deep learning techniques and was trained on a significant dataset of human positions [8, 9]. It has a lightweight design that enables real-time operation on embedded and mobile devices, making it appropriate for a variety of applications. Fig. 1 shows the different body landmarks used in the MediaPipe model.

A robust and adaptable pose-tracking model, MediaPipe Pose enables various applications in several industries. It is a valuable tool for developers and designers who want to build interactive and immersive user experiences because of its speed, precision, and versatility. In this research, we utilized the MediaPipe Pose model to track upper body motion, which controls the robot manipulator. MediaPipe is preferred over competing posture-tracking frameworks for many reasons, including its real-time speed optimization, effective use of GPU and CPU resources, and ability to handle various situations and positions.

BlazePose does pose estimation in two steps. In the first stage, a detector finds the posture region of interest (ROI) in the video frame. The tracker predicts all 33 posture key points inside this ROI in the second phase. BlazePose examines the first frame to pinpoint critical moments. The system comprises two machine learning models: The Detector, which isolates the human region from the input picture, and the Estimator, which takes a 256×256 px image of the person and returns key points.

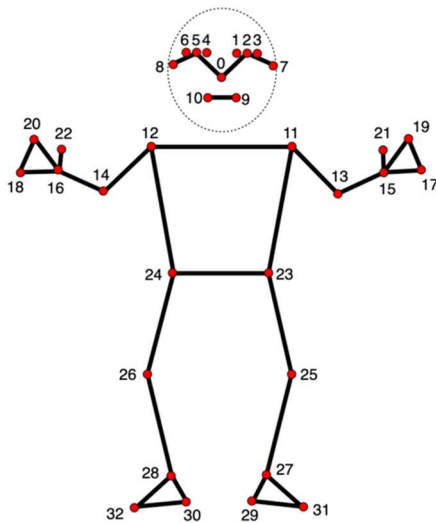


Fig. 1. Body landmarks of the BlazePose model.

B. Related Works

A robotic device that focuses on gesture motions and makes use of a wireless communication system is shown in the paper by Mahesh and Singh [10]. This device's design reportedly outperforms previous remote-controlled robots regarding security and user-friendliness. Input for human-computer interaction is provided through a wired glove. High degrees of repeatability, accuracy, and reliability, essential for robotic applications, are used by the glove's sensors to gather physical data, such as the bending of fingers.

Research on HRI that employs robot teleoperation and vision is intimately tied to the advancement of artificial intelligence. Xu et al. [11] suggested that this issue was addressed by robot teleoperation and human-robot interface techniques based on digital twin technology. The slave robot comprises a Pisa/IIT SoftHand-equipped, lightweight robotic arm from KUKA that is controlled in impedance to carry out a drilling operation—an illustration of a dynamic activity with environmental restrictions.

Zhao et al.'s [12] proposal for an intuitive human-robot interface for real-time teleoperation allows human operator control of the Baxter robot to carry out challenging tasks in an unstructured and unpredictable environment. They extracted body motion using the Kinect sensor and mapped the workspace using inverse kinematics. A method for human motion capture based on the Kinect and IMUs and its application to HRI was proposed by Zhang et al. [13]. To compensate for the accuracy of human posture capture under dynamic, magnetically disturbed, and occlusion settings, their proposed solution integrated the features of Kinect data with high precision and IMU data with stability. A real-time remote robot teleoperation approach employing markerless Kinect-based hand tracking was presented by Du et al. [14]. By analyzing depth data from the Kinect, this tracking technique may be used to predict the 3D locations of the index finger and thumb. The hand stance is employed as a model to specify the end-effector pose of a real-time remote robot.

A modular system for untethered real-time kinematic motion capture employing sensors with inertial measuring units (IMU) was developed by Miller et al. [15]. A collection of compact and lightweight sensors made up their system. Additionally, they presented the outcomes of teleoperating the NASA Robonaut using a motion capture system in real-time. Li et al.'s [16] proposal for manipulating a robot arm was based on the evaluated algorithms of gesture identification and limb movement recognition in somatosensory contact. To be more precise, gesture and arm movement are located and identified using 3D SSD architecture. A unique multi-sensor guided hand gesture detection system for surgical robot teleoperation was proposed by Qi et al. [17]. A multi-sensor data fusion model has been developed to conduct interference in the presence of occlusion. A multilayer Recurrent Neural Network (RNN) consisting of an LSTM module and a dropout layer has been proposed to categorize numerous hand gestures.

III. METHODOLOGY

This section discusses the design and development of the proposed system. Two sides comprise the proposed

methodology: one is for the operator, and the other is for the robot manipulator. A camera video stream was acquired on the user's side, and the user's right-hand landmarks were recorded. These landmarks were used to compute the elbow and shoulder angles, which were then used to derive the theoretical x and y values using forward kinematics. When the robot manipulator side receives this information wirelessly and controls the joint angles accordingly, the motors that drive the manipulator on the robot are then provided with the angle data. The device allows the operator to watch a live video of the robot manipulator environment. This technology enables operators to manage robot manipulators naturally and straightforwardly, making them appropriate for various tasks. Fig. 2 shows the overall architecture of the proposed system.

A. Development of Robot Manipulator and Controller

This section discusses the design and development of a 3-DoF robot manipulator. Fig. 3 depicts the joint and link architecture of the proposed robot manipulator. The robot's base and joints used metal gear servo motors with lever controls. Every component, including the base, brackets, and gripper, was 3D printed. For improved joint-servo interactions and effective operation, gear horns were incorporated.

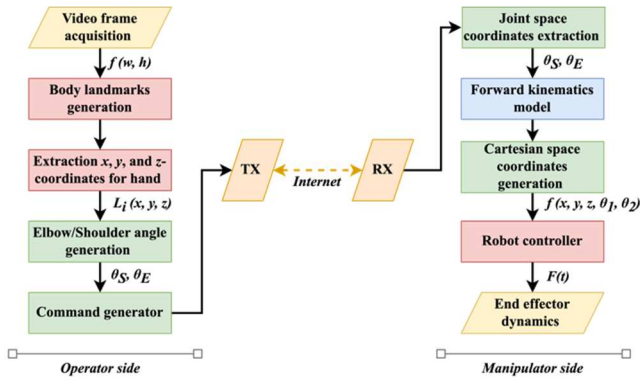


Fig. 2. Architecture of the user side-robot side control mechanism.

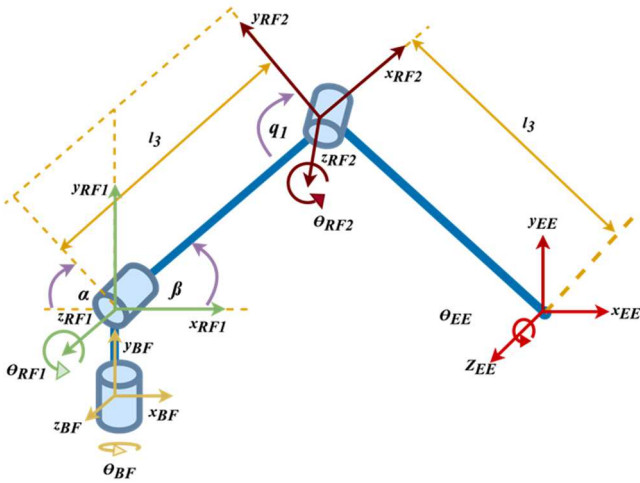


Fig. 3. Joint and link architecture of the proposed robot manipulator.

Fig. 4 depicts the components of the control box for the robot manipulator, including the buck converter, motor driver, Raspberry Pi, touch display, and power supply. A 12V/20A power supply was utilized to power the robot manipulator, and a buck converter was used to step down to 5V. A 16-Ch 12-bit PWM I2C servo motor driver module with a DC 5–10V voltage requirement was used to drive three servo motors. The Raspberry Pi used an I2C connection to interact with the servo motor driver module.

B. Upper Body Motion Tracking and Angle Calculation

Body movements must determine the user's elbow and shoulder angles. To calculate one angle, we utilized three landmarks. For instance, we used the hip, shoulder, and elbow landmark positions to compute the shoulder angle. Fig. 5 shows the landmarks of the hip, shoulder, elbow, and wrist joints. The first stage was finding the three landmarks and figuring out their x and y coordinates. The methods described in the Numpy module were then used to compute the angles and convert them to degree format. Consequently, we performed the same step to compute the elbow angle with the aid of the x and y -coordinates of the shoulder, elbow, and wrist.

A geometrical approach is needed to determine a point's unknown coordinate or the distance between two points in Cartesian space. Fig. 6 shows the angle calculation methodology using the proposed geometrical method. Equation (1) is the distance formula for finding the distance between two points in a two-dimensional Cartesian coordinate system. Equation (2) represents the Law of Cosines, which is used to determine the measure of an angle in a triangle when the lengths of its sides are known.

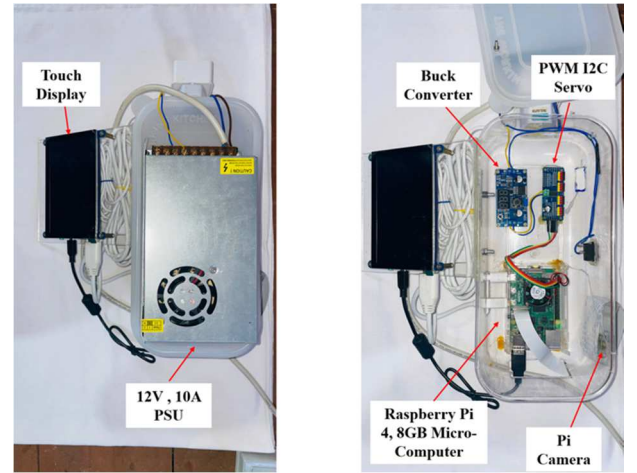


Fig. 4. Hardware components of the robot controller.

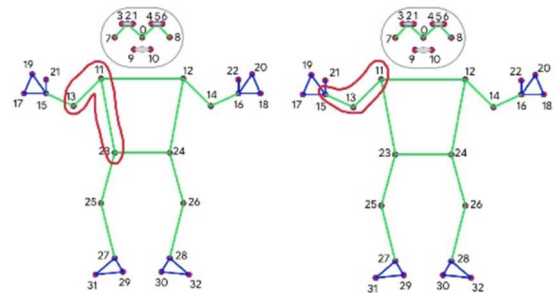


Fig. 5. BlazePose landmarks of the hip, shoulder, elbow, and wrist joints.

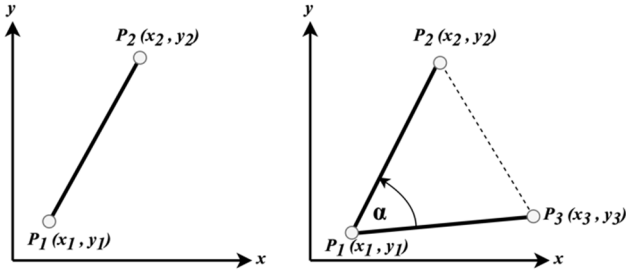


Fig. 6. Joint angle calculation methodology.

$$P_1P_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (1)$$

$$\alpha = \cos^{-1} \left(\frac{P_1P_2^2 + P_1P_3^2 - P_2P_3^2}{2(P_1P_2 \times P_1P_3)} \right) \quad (2)$$

C. Mapping Human-Robot Interaction and Realtime Controlling

Fig. 7 shows the joint and link motion mapping of the proposed robot arm and human hand. The obtained kinematics variables aided the geometrical shape represented in the illustration. Fig. 7(a) shows that we got the theoretical Cartesian coordinates corresponding to the hand above motion angles. Equation (3) and (4) describes the Cartesian coordinates of the human hand. The variables X and Y represent the position of the end of the hand in the x - y coordinate system. Where l_1 and l_2 represent the length between wrist to elbow and elbow to the shoulder of the human hand.

$$X = l_1 \sin(\theta_S) + l_2 \sin(\theta_E - \theta_S) \quad (3)$$

$$Y = l_2 \cos(\theta_E - \theta_S) - l_1 \cos(\theta_S) \quad (4)$$

As shown in Fig. 7(b), the angles were sent to the robot arm to move the end-effector to the desired location determined by X and Y coordinates in the previous phase. Then, using the results, produce the joint angles of the manipulator, as described in eq. (5) and (6). l_3 is the length of the link (80 mm) of the robot. The β angle will transfer to joint 1 while the α angle (described in eq. (7)) will transfer to joint 2 of the robot.

$$q_1 = \cos^{-1} \left(\frac{X^2 + Y^2 - 2l_3^2}{2l_3^2} \right) \quad (5)$$

$$\beta = \tan^{-1} \left(\frac{Y}{X} \right) + \tan^{-1} \left(\frac{l_3 \sin(q_1)}{l_3 + l_3 \cos(q_1)} \right) \quad (6)$$

$$\alpha = q_1 - \beta \quad (7)$$

The system was remotely controlled via LAN (Local Area Network) technology. The TCP (Transmission Control Protocol) Socket offered a connection-oriented, reliable, stream-based communication. Due to threading, real-time operation was feasible when both sides ran the code and connected to the network.

IV. RESULTS AND ANALYSIS

This section discusses the results and analyses of the user's upper body motion tracking, robot manipulator kinematics, and, task space mapping. The following three steps were used as the experimental scenario of the research: landmark detection, angle calculation, and communication between a person and a robot.

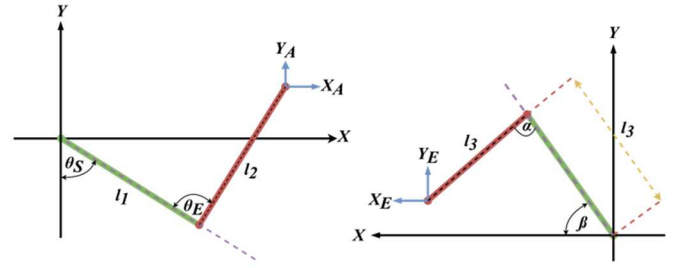


Fig. 7. Joint and link motion mapping, (a: Left) human hand, (b: Right) robot arm.

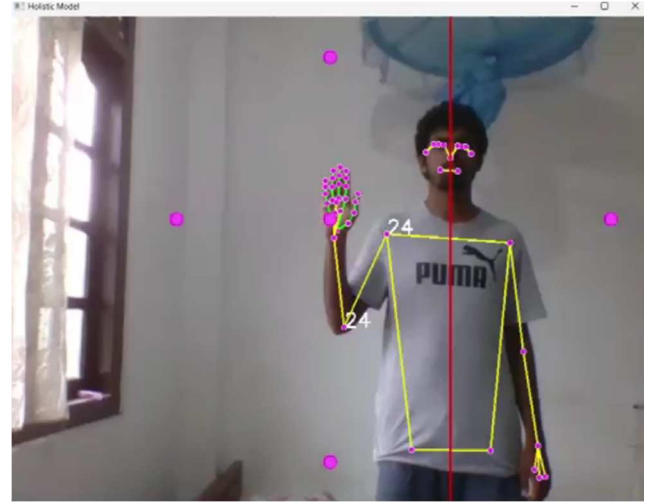


Fig. 8. Upper body motion tracking experiment.

TABLE I. EXPERIMENT RESULTS ON UPPER BODY MOTION TRACKING ON DIFFERENT TARGET POINTS (WHERE θ_S =SHOULDER ANGLE, θ_E =ELBOW ANGLE)

#	Up / (°)		Down / (°)		Middle / (°)		Side / (°)	
	θ_S	θ_E	θ_S	θ_E	θ_S	θ_E	θ_S	θ_E
1	158	162	12	172	23	17	90	170
2	170	167	11	178	29	22	90	173
3	160	165	10	178	25	10	93	170
4	165	170	13	175	35	30	90	165
5	171	168	11	171	30	25	85	168

Fig. 8 depicts the experimental setup used to do tests on landmark detection and angle computations. The experiment comprises five target points (pink dots), and the natural center of the human body (red line). The user maintains the right hand on each target point for five seconds in the following order: up, down, middle, then side (user's right side). Users must always maintain their body centred or close to the red line.

After moving the user's hands and wrists to the center of the target points while standing close to the red line, we observed and recorded the angles presented in the video stream. The shoulder and elbow angles were recorded from 40 (M : 55%, F : 45%) individuals during five sessions (200 trials) that were carried out comparably. Five randomly chosen individuals' test results are shown in Tab. 1.

Fig. 9 shows how various target sites' shoulder and elbow angles vary. The many data points for multiple test situations and test participants revealed the same fluctuation, as seen in the illustration. This finding indicated that our upper body

motion-tracking technology produced positive outcomes. A confusion matrix was used to record and analyze the trial data from 40 people. According to the confusion matrix data, the upper-body motion tracking system categorized four different motion classes with an accuracy rate of 96.5%. There is a 2.0% fluctuation in the findings, which we also noticed. It is also evident that the functioning and reaction curve changed as people aged.

The difference in lighting conditions is the main cause of the variances shown in the curves (see Fig. 9). In such cases, the algorithm occasionally has trouble correctly determining the locations of the objects. Notably, the accuracy attained in this vision-based technique is noticeably higher when compared to other methods that depend on tangible objects or wearables.

A basic 3D printed arm with 3-DoF and two links allowing bidirectional movement is shown as a robot manipulator in Fig. 10. The manipulator's base has two motors controlled by two lever mechanisms. The first and second motors, respectively, control the movement of the first and second joints, while the third motor controls the rotation of the manipulator. The manipulator can be moved several ways by varying the motors' speed and direction.

Tab. 2 depicts the results of the system for each of the 20 test cases. The vision system analyzed two input parameters: elbow and shoulder angles. As an output from the robot, we observed the rotational axes of the base, joint 1 and joint 2. The mapping produced accurate results with 90% success.

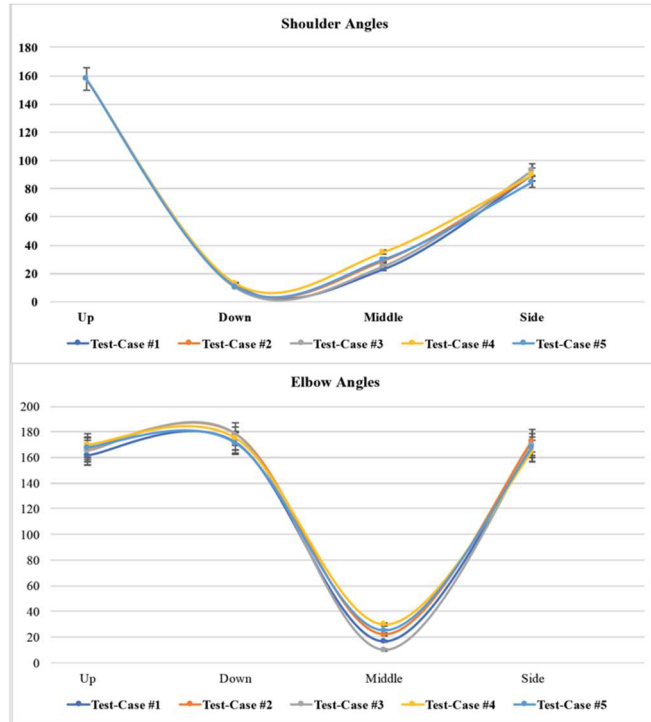


Fig. 9. Angle variation over different target points, (Top: shoulder angles, Bottom: elbow angle).

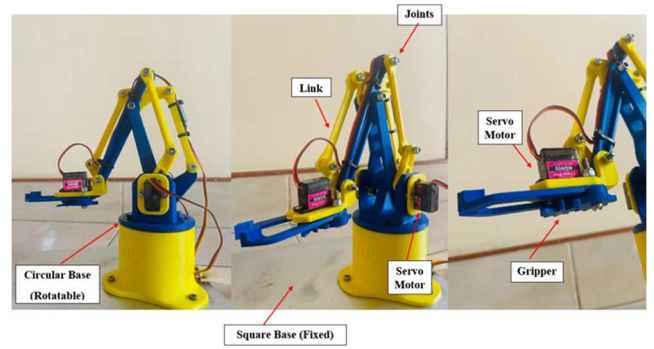


Fig. 10. Physical view of the 3-DoF robot manipulator.

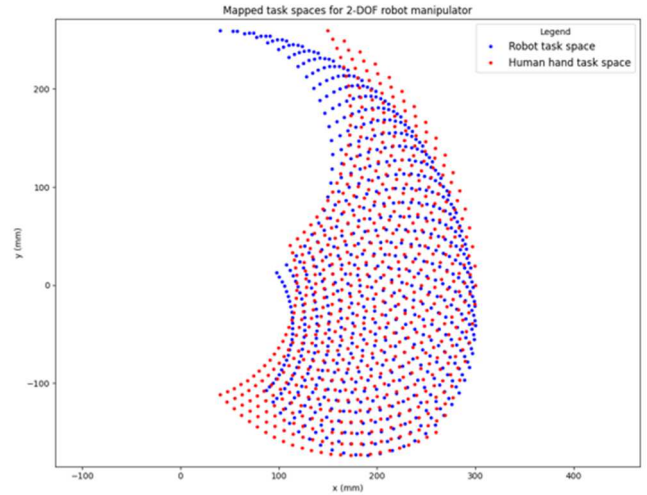


Fig. 11. Task space comparison, (Red: human, Blue: robot).

TABLE II. EXPERIMENT RESULTS (WHERE θ_s =SHOULDER ANGLE, θ_e =ELBOW ANGLE, β = ROBOT JOINT 1, α = ROBOT JOINT 2, ϕ =BASE ORIENTATION)

#	$\theta_s / (^{\circ})$	$\theta_e / (^{\circ})$	$\beta / (^{\circ})$	$\alpha / (^{\circ})$	ϕ
1	20	180	60	100	0.79
2	65	150	29	132	0.7
3	40	48	98	74	0.53
4	66	110	46	118	0.48
5	73	123	33	130	0.5
6	25	13	90	76	0.43
7	25	100	90	74	0.33
8	36	136	64	98	0.28
9	42	90	90	79	0.57
10	68	70	62	107	0.65
11	70	185	9	154	0.48
12	55	36	87	87	0.57
13	24	78	90	65	0.26
14	10	15	90	56	0.18
15	79	108	35	131	0.68
16	90	104	26	140	0.57
17	66	55	70	102	0.75
18	78	135	24	140	0.2
19	67	124	40	125	0.16

#	$\theta_s / (^{\circ})$	$\theta_e / (^{\circ})$	$\beta / (^{\circ})$	$\alpha / (^{\circ})$	Θ
20	45	88	86	80	0.55

Fig. 11 illustrates how robots can partially penetrate the human work environment. We found that a robot can go beyond the bounds of human task space. However, we also considered against using the margin because it was the robot's safety margin.

V. CONCLUSION

In conclusion, real-time upper body motion tracking using computer vision algorithms is an intriguing field of research with great potential for enhancing teleoperation and human-robot interaction. Such a system can improve the accuracy, efficiency, and intuitiveness of the robot's motions and enable more natural and efficient collaboration between people and robots by giving real-time input to a robot based on the upper body movements of a human operator. We exploited the user's upper body motion to move the robot manipulator by utilizing the Google MediaPipe pose tracing model and appropriately derived robot joint angles. While the overall system accuracy was 94.1 (± 1.2) %, the vision system's body motion tracking accuracy was 96.5 (± 2.0) %.

Increasing the smoothness of the robot's motions by increasing the number of target points is one potential area of focus for studying real-time upper body motion tracking using computer vision. Integrating AI methods into the system to provide more accurate task space mapping values is another exciting avenue. This may entail employing machine learning algorithms to examine and gain knowledge from the data gathered by the computer vision system to provide more accurate and trustworthy estimations of the human operator's upper body motions.

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